

Hassan Mahdi Hassan Diab

AI Computer Vision Engineer | Associate Software Engineer | Full-Stack Developer

✉ hassanmahdi102002@gmail.com ☎ 01034230912 📍 Zagazig 🌐 Hassan Mahdi

👤 Hassan Mahdi 📁 Portfolio

SUMMARY

Computer Engineering graduate with a strong foundation in Artificial Intelligence, Computer Vision, and Full-Stack Development. Demonstrated ability to independently architect and develop end-to-end intelligent systems, from NoSQL backends to AI-driven diagnostic tools. Passionate about solving complex technical challenges and delivering robust, scalable applications using modern technologies.

EDUCATION

BSc. in Computer Engineering and technology

06/2026

Modern Academy for Engineering and Technology | Graduated: 2026

Maadi, Egypt

(Grade: Very Good - GPA: 2.922)

Graduation Project: Smart Dental Diagnostic System Using AI.

- Description: A multi-stage computer vision pipeline for dental lesion localization and disease classification.

SKILLS

Artificial Intelligence & Computer Vision:

Deep Learning (CNNs), Object Detection, Multi-Label-Classification, Conversational AI, Chatbots, Natural Language Processing (NLP), LLM Integration, Prompt Engineering.

Backend & Database Architecture:

RESTful API Design and Development, Node.js, Express.js, NoSQL Database Design, MongoDB, Secure Data Management, End-to-End System Integration.

Frontend Development:

React.js, HTML5, CSS3, JavaScript, Tailwind CSS, Bootstrap.

Programming Languages:

Python (AI), C++, JavaScript (Web).

Tools, Concepts & Libraries:

Git/GitHub (Version Control), Postman (API Testing), Object-Oriented Programming (OOP), Data Structures.

Professional Skills:

End-to-End System Ownership, Autonomous Problem Solving, Technical Crisis Management, Software Architecture Planning, Adaptability.

LANGUAGES

Arabic

Native

English

Fluent

PROJECTS

Smart Dental Diagnostic system using AI

08/2025 – 06/2026

Lead AI | Backend Developer | Frontend Developer | Graduation Project

- Developed an end-to-end intelligent diagnostic mobile application capable of identifying eight distinct oral conditions using color intraoral smartphone images.
- Solely architected, trained, and implemented the core Computer Vision models from scratch utilizing **YOLOv8** and **ResNet50** for precise image analysis.
- Engineered an advanced, interactive **AI Agent (Chatbot)** powered by the Gemini API, functioning as a smart virtual dental assistant to facilitate real-time user consultations and answer complex medical queries.
- Integrated the **conversational AI** Agent to **autonomously** analyze diagnostic results and generate comprehensive, **context-aware** medical reports.
- Designed and built a scalable backend infrastructure independently utilizing a NoSQL architecture (**MongoDB**), ensuring secure and efficient medical data management.

Around the World - Countries Explorer | Personal Project

06/2026 – 07/2026

Frontend Developer

- Developed a dynamic, fully responsive Single Page Application (SPA) using **React.js** and **Tailwind CSS**, integrating the REST Countries API to display comprehensive global data.
- Implemented advanced search functionality, region-based filtering, and a custom useFetchData hook with **LocalStorage** caching to optimize performance and reduce API load.

Product Management System (CRUD)

05/2026 – 05/2026

Frontend Developer

- Built a web application for managing product inventories supporting full CRUD operations, dynamic search, and robust form validation using Regex and LocalStorage for data persistence.

Interactive Calculator

08/2026 – 08/2026

Frontend Developer

- *Designed a fully functional calculator with a modern UI utilizing CSS Grid and dynamic CSS variables.*
- Implemented basic arithmetic operations, a light/dark theme toggle, and full keyboard accessibility for seamless user interaction using Vanilla JavaScript.